Lab Manual: Practical 9

Course Code: ETCS406

Course Title: Game Development

Program: B. Tech Computer Science and Engineering

Semester: VII

Practical Title: Final Mini-Project – Develop a Complete Game Prototype in Unity (2D/3D/AR/VR)

Duration: 10 Hours

Lab Type: Capstone Project

Assessment Weightage: CAL 3 (Presentation on Mini Project) + Term End Oral Examination

Objective

To design, develop, and present a complete game prototype using Unity, integrating core gameplay mechanics, user interface, sound, and deployment features. This mini-project serves as a culmination of all skills learned throughout the course.

Pre-requisites

- Proficiency in Unity Editor and C# scripting
- Understanding of game design principles (MDA framework, level design, narrative)
- Familiarity with physics, animation, UI systems, and asset integration
- Experience with AR/VR tools (if applicable)

Learning Outcomes

By the end of this lab, students will be able to:

- Apply comprehensive game development skills to build a playable prototype
- Integrate multiple Unity features including UI, sound, physics, and animation
- Design immersive and engaging gameplay experiences
- Deploy and present the game with supporting documentation

Tools & Resources

- Unity Editor (latest stable version)
- Visual Studio or preferred C# IDE
- Unity Asset Store (for free or custom assets)
- XR Interaction Toolkit (for AR/VR projects)
- Audio editing tools (e.g., Audacity)
- Screen recording software (e.g., OBS Studio)

Lab Tasks

Task 1: Game Concept Finalization

- Choose game type: 2D, 3D, AR, or VR
- Define genre, target audience, platform, and core mechanics
- Create a design document outlining gameplay, levels, UI, and narrative

Task 2: Development Phase

- Set up Unity project and organize assets
- Implement core mechanics (movement, interaction, scoring, etc.)
- Design levels or environments using tilemaps, terrain, or prefabs
- Integrate animations, sound effects, and background music
- Build UI elements (menus, HUD, feedback systems)

Task 3: Advanced Features (Optional)

- Add AI components (e.g., NPC behavior, pathfinding)
- Include AR/VR interactions using XR Toolkit
- Implement save/load functionality or player progression

Task 4: Testing & Debugging

- Conduct internal playtesting and fix bugs
- Optimize performance and polish visuals
- Ensure cross-platform compatibility (PC, mobile, headset)

Task 5: Deployment & Presentation

- Build executable file or APK
- Record gameplay video showcasing features

- Prepare functional documentation including:
    - Game overview
    - Features and mechanics
    - Screenshots and flow diagrams
    - Challenges faced and solutions
- Present the project during oral examination

Evaluation Criteria

Component	**Marks**
Game Design & Concept	5
Functionality & Mechanics	5
UI/UX & Visual Polish	5
Documentation & Presentation	5
Oral Examination (Viva)	25
Total	**45**

Tips for Success

- Keep scope manageable—focus on polish over complexity
- Use version control (e.g., Git) to track progress
- Test on actual devices if deploying to mobile or AR/VR
- Make your presentation engaging with visuals and a live demo

Suggested Reading & Resources

- *The Art of Game Design* by Jesse Schell – Final chapters on production and pitching
- Unity Learn: Game Development Pathways
- Gamasutra: Postmortems and indie dev insights
- YouTube: Devlogs and Unity showcase channels

Submission Guidelines:

- Submit your report in PDF format via the Google Classroom.
- Include screenshots or gameplay references where relevant.
- Submit a video of your game play.